



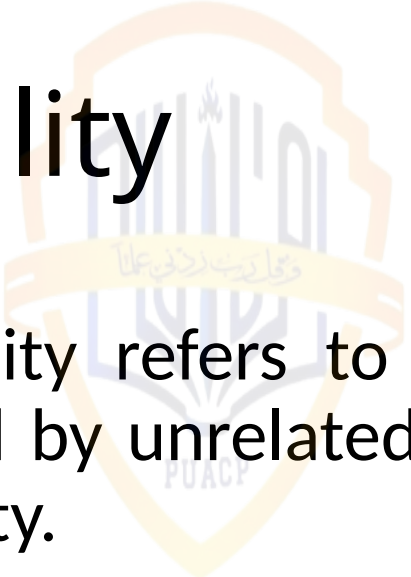
Externalities and Public Goods

Outlines

- Externality



Externality



- An externality refers to the cost or benefit of an activity that is experienced by unrelated third parties who are not directly involved in the activity.
- In other words, externalities occur when the actions of individuals or firms have spill-over effects on others in society, either imposing costs or conferring benefits without compensation.
- These external effects are not reflected in the market prices of goods or services involved in the activity, leading to market failures and inefficiencies.

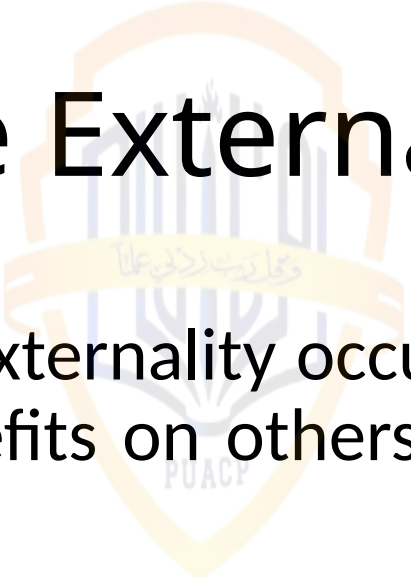
Examples of Externalities

- Examples of externalities include pollution from factories affecting nearby communities (negative externality) or vaccinations reducing the risk of disease transmission for the broader population (positive externality).

Negative Externality

- A negative externality occurs when the actions of an individual or firm impose costs on others in society who are not directly involved in the activity.
- Example: Pollution emitted by a factory negatively impacts the health of nearby residents and damages the environment, imposing costs on them without compensation.
- Impact: Negative externalities lead to overproduction or consumption of goods or services that generate harmful effects, resulting in a misallocation of resources and a reduction in societal welfare.

Positive Externality

The logo of the Punjab University of Agriculture, Chokkhi, Pakistan (PUACP) is visible in the background. It features a shield-like emblem with a book and a torch, and the university's name in Urdu and English.

- A positive externality occurs when the actions of an individual or firm confer benefits on others in society who are not directly involved in the activity.
- Example: Education benefits not only the individual student but also society as a whole by increasing productivity, innovation, and social cohesion.
- Impact: Positive externalities lead to underproduction or consumption of goods or services that generate beneficial effects, resulting in a suboptimal allocation of resources and a missed opportunity to enhance societal welfare.

Consumption Externality

- **Definition:** A consumption externality occurs when the consumption of a good or service by one individual directly affects the utility or well-being of other individuals in society.
- **Example:** Secondhand smoke from cigarette consumption negatively affects the health and well-being of non-smokers who are exposed to it involuntarily.

Example of Positive Externality

- In a neighborhood, if one neighbor has a pretty garden, it can make the whole area look nicer. Even though the neighbor takes care of the garden for their own enjoyment, everyone else who walks by or lives nearby also gets to enjoy it. This makes the neighborhood more attractive and can even raise the value of the houses around it.
- So, the pretty garden creates a good feeling for everyone in the neighborhood, even if they didn't do anything to make it happen.

Consumption Externality

- **Impact:** Consumption externalities lead to a divergence between private and social benefits of consumption. If the externality is negative (e.g., pollution), individuals may consume more than the socially optimal level, leading to a misallocation of resources and reduced societal welfare. If the externality is positive (e.g., vaccination), individuals may consume less than the socially optimal level, leading to underprovision of the beneficial good or service.

Production Externality

- **Definition:** A production externality occurs when the production of a good or service by one firm affects the costs or benefits of other firms or individuals in society.
- **Example:** A factory emitting pollutants into the air or water imposes negative production externalities on nearby residents and ecosystems by degrading air quality and harming the environment.

Positive Production Externality

- One classic example of positive production externality is the relationship between an apple orchard and nearby beekeepers. In this scenario, the beekeepers place their hives near the apple orchard to allow their bees to forage for nectar. As the bees collect nectar from the apple blossoms, they inadvertently transfer pollen between flowers, aiding in the pollination process. This increased pollination results in higher fruit yields and improved fruit quality for the apple orchard. While the beekeepers' primary aim may be to produce honey, their bees provide a valuable service to the orchard by enhancing its productivity.
- This example illustrates how two seemingly unrelated activities can benefit each other through positive externalities.

Production Externality

- **Impact:** Production externalities lead to a divergence between private and social costs of production. If the externality is negative (e.g., pollution), firms may produce more than the socially optimal level, leading to overproduction of goods or services with harmful effects on society. If the externality is positive (e.g., research and development), firms may produce less than the socially optimal level, leading to underinvestment in activities with beneficial spillover effects.

Key terms to remember

- **Marginal Private Cost (MPC):**
- **Marginal External Cost (MEC):**
- **Marginal Social Cost (MSC):**
- $MSC = MPC + MEC$
- **Marginal Private Benefit (MPB):**
- **Marginal External Benefit (MEB):**
- **Marginal Social Benefit (MSB):**
- Formula: $MSB = MPB + MEB$

Marginal Private Cost (MPC):

- Definition: Marginal private cost refers to the additional cost incurred by an individual or firm for producing one more unit of a good or service, taking into account only the private costs directly borne by the producer.
- Example: For a manufacturing company, the marginal private cost includes expenses such as raw materials, labor, and equipment needed to produce an additional unit of the product.

Marginal External Cost (MEC):

- Definition: Marginal external cost refers to the additional cost imposed on third parties or society as a whole for producing one more unit of a good or service, beyond the private costs borne by the producer. It represents the negative externality associated with the production process.
- Example: In the case of a factory emitting pollution into the air, the marginal external cost includes the health and environmental damages caused by the pollution, which are not directly borne by the producer but are incurred by society.

Marginal Social Cost (MSC):

- Definition: Marginal social cost represents the total cost to society of producing one more unit of a good or service, taking into account both the private costs borne by the producer and the external costs imposed on third parties or society as a whole.
- Formula: $MSC = MPC + MEC$
- Example: When calculating the marginal social cost of production, both the private costs (MPC) and the external costs (MEC) associated with the production process are considered to determine the full societal cost of producing an additional unit of the good or service.

Marginal Private Benefit (MPB)

- Definition: Marginal private benefit refers to the additional benefit received by an individual or firm from consuming one more unit of a good or service, taking into account only the private benefits directly enjoyed by the consumer.
- Example: For a consumer purchasing a smartphone, the marginal private benefit includes the satisfaction, convenience, and utility gained from using the device for personal purposes.

Marginal External Benefit (MEB)

- Definition: Marginal external benefit refers to the additional benefit enjoyed by third parties or society as a whole from the consumption of one more unit of a good or service, beyond the private benefits enjoyed by the consumer. It represents the positive externality associated with consumption.
- Example: When a homeowner installs solar panels on their roof, the marginal external benefit includes the reduction in air pollution and greenhouse gas emissions, as well as the contribution to renewable energy generation, which benefit the community and the environment.

Marginal Social Benefit (MSB):

- Definition: Marginal social benefit represents the total benefit to society from consuming one more unit of a good or service, taking into account both the private benefits enjoyed by the consumer and the external benefits enjoyed by third parties or society as a whole.
- Formula: $MSB = MPB + MEB$
- Example: When calculating the marginal social benefit of consumption, both the private benefits (MPB) and the external benefits (MEB) associated with the consumption activity are considered to determine the full societal benefit of consuming an additional unit of the good or service.